

CYBERSECURITY THREAT
REPORT TASK-1
KODURU NIDESH

INTRODUCTION

- Introduction to cybersecurity
- Identify 5 Major Modern Cyber Threats
- Impact Analysis
- Real- World Case Studies
- Preventive Measures
- Conclusion & Future Scope

What is Cyber security

Cybersecurity is the practice of protecting computers, networks, devices, and digital data from cyberattacks, unauthorized access, theft, and damage. It involves technologies, processes, and security measures that help keep information safe and confidential.

- Protects personal and business data from hackers. Prevents financial losses caused by cybercrime.
- Ensures privacy and secure online transactions.
- Maintains business continuity and customer trust.
- Protects critical systems from cyber threats.

Why is Cybersecurity Important for Individuals and Businesses?

- **For Individuals**

- Protects personal information such as passwords, bank details, and identity data.
- Prevents online fraud, identity theft, and financial losses.
- Ensures privacy while using the internet and digital services.

- **For Businesses**

- Protects sensitive company and customer data.
- Prevents financial losses caused by cyberattacks and data breaches.
- Maintains business operations and reduces downtime.
- Builds customer trust and protects the organization's reputation.
- Helps comply with legal and regulatory requirements.

AI-Powered Phishing Attacks :

AI-Powered Phishing is a cyberattack where criminals use Artificial Intelligence (AI) to create highly convincing emails, messages, voice recordings, or deepfake videos to trick people into revealing sensitive information.

- In 2024, cybercriminals used AI-generated deepfake videos and voice clones to impersonate company executives and deceive employees into transferring large amounts of money.

Ransomware-as-a-Service (RaaS):

Ransomware-as-a-Service (RaaS) is a cybercrime model where ransomware developers sell or rent ransomware tools to criminals, allowing them to launch attacks without advanced technical skills.

- WannaCry Ransomware Attack (2017) – A global ransomware attack that infected over 200,000 computers across 150 countries, disrupting hospitals, businesses, and government organizations.

Cloud Security Misconfigurations:

Cloud Security Misconfigurations occur when cloud services such as AWS, Google Cloud Platform (GCP), or Microsoft Azure are configured incorrectly, exposing sensitive data and systems to unauthorized access.

- Capital One Data Breach (2)019 – A cloud configuration vulnerability exposed the personal information of over 100 million customers, making it one of the largest cloud-related data breaches.

IoT Vulnerabilities (Smart Devices Being Hacked):

IoT (Internet of Things) Vulnerabilities are security weaknesses in smart devices such as smart cameras, smart TVs, smart speakers, fitness trackers, and home automation systems that can be exploited by hackers.

Impact Analysis

Threat	Impact on Individuals	Impact on Organizations
AI-Powered Phishing	Identity theft, financial fraud, stolen credentials	Data breaches, financial losses, reputation damage
Ransomware-as-a-Service (RaaS)	Loss of personal files, financial loss	Operational downtime, data loss, ransom payments, compliance issues
Cloud Security Misconfigurations	Exposure of personal information, privacy breaches	Data leaks, regulatory penalties, loss of customer trust
IoT Vulnerabilities	Privacy invasion, unauthorized access to smart devices	Network compromise, data theft, business disruption

Key Impacts on Organization:

- Loss of sensitive business data
- Operational downtime and productivity loss
- Reputation and customer trust damage
- Legal, regulatory, and compliance issues

Key Impacts on Individuals:

- Data theft and identity theft
- Financial fraud and monetary loss
- Privacy breaches
- Unauthorized access to personal accounts

Real-World Case Studies

AI-Powered Phishing Attacks – Hong Kong Deepfake Scam (2024)

In 2024, a multinational company in Hong Kong became the victim of an AI-powered phishing attack.

Cybercriminals used deepfake technology to create realistic video and audio impersonations of senior executives during a virtual meeting. Believing the instructions were legitimate, an employee transferred approximately \$25 million to fraudulent accounts.

This incident demonstrated how artificial intelligence can be used to make phishing attacks more convincing and difficult to detect.

Ransomware-as-a-Service (RaaS) – WannaCry Ransomware Attack (2017)

The WannaCry ransomware attack was one of the largest cyberattacks in history, affecting more than 200,000 computers across 150 countries. The malware exploited a vulnerability in Microsoft Windows and encrypted victims' files, demanding ransom payments in Bitcoin to restore access. The attack severely disrupted hospitals, businesses, and government agencies, causing billions of dollars in damages and highlighting the importance of timely software updates and backups.

Cloud Security Misconfigurations – Capital One Data Breach (2019):

In 2019, Capital One suffered a major data breach due to a cloud security misconfiguration. An attacker exploited a vulnerability in the company's cloud infrastructure and gained access to sensitive information belonging to over 100 million customers. The exposed data included names, addresses, credit scores, and other personal details. This incident emphasized the need for proper cloud configuration, continuous monitoring, and strong access controls.

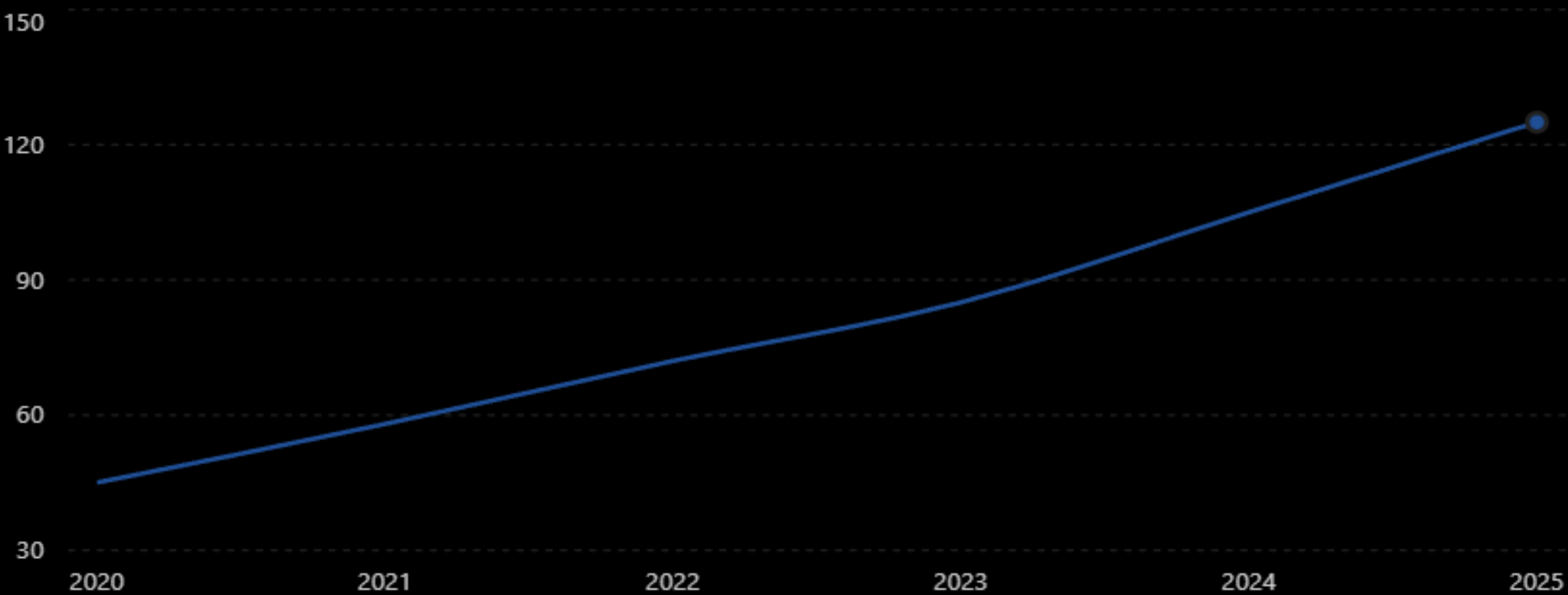
Preventive Measures

- **1. AI-Powered Phishing Attacks**
- Implement Multi-Factor Authentication (MFA).
- Conduct regular Security Awareness Training.
- Use advanced email filtering and anti-phishing tools.
- **2. Ransomware-as-a-Service (RaaS)**
- Perform regular data backups.
- Apply regular Patch Management.
- **3. IoT Vulnerabilities**
- Deploy Endpoint Protection and Anti-Ransomware solutions.
- **4. Cloud Security Misconfigurations**
- Implement strong Identity and Access Management (IAM).
- Conduct regular cloud security audits.
- Encrypt sensitive data in storage and transit.
- Change default usernames and passwords.
- Regularly update device firmware and software.
- Use network segmentation to isolate IoT devices.

Growth of Major Cybersecurity Threats (2020–2025)



Illustrative trend showing the increasing prevalence of major cyber threats in recent years.



Conclusion & Future Scope

- **Conclusion:**

Cybersecurity is essential for protecting individuals, businesses, and critical systems from modern cyber threats. Proactive security measures such as Multi-Factor Authentication (MFA), regular patch management, security awareness training, and continuous monitoring help prevent cyberattacks, reduce risks, and protect sensitive information.

- By implementing strong security practices, organizations can maintain business continuity, avoid financial losses, and preserve customer trust.

- **Future Scope:**

As technology continues to advance, cyber threats are becoming more sophisticated through Artificial Intelligence, cloud computing, and connected devices. Therefore, continuous learning and staying updated with the latest cybersecurity trends are crucial for both individuals and organizations.

- Future cybersecurity efforts will focus on AI-driven defense systems, Zero Trust Security, cloud security automation, and advanced threat intelligence to combat evolving cyber risks and ensure a secure digital environment.

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